

# Tao Wang

Doctoral Candidate

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## Education

- **Ph.D.**, Cyberspace Security, Nanjing University of Aeronautics and Astronautics 2024.04–2027.04
- **Visiting Ph.D.**, Cyberspace Security, University of Milan, SPDP Lab 2025.09–2026.09
- **M.S.**, Cyberspace Security, Nanjing University of Aeronautics and Astronautics 2021.09–2024.04
- **B.E.**, Computer Science and Technology, Anhui Normal University 2017.09–2021.06

## Research Interests

- **Field:** Data privacy, computational social science, and artificial intelligence
- **Focus:** Implicit privacy, usable privacy, visual privacy, and privacy evaluation

**Previous Research:** Controllable Identity Visibility in Usable Face Privacy:

- Universal Privacy Protection: An information-theoretic face anonymization model that removes identity information from both human and machine observers [C1].
- Socially-Aware Privacy: A de-identification approach that preserves recognizability for familiar individuals while preventing machine recognition [J8].
- Human–AI Privacy Asymmetry: An identity hider that leverages differences between human and machine perception, only enabling machine utility [J9].
- Authorized Identity Access: A cancelable face generation model that supports identity recognition exclusively by authorized systems [C3].

**Current and Future Research:** Human-Centered Implicit Privacy Governance in the GenAI Era.

- As GenAI systems like GPT increasingly infer sensitive attributes, intentions, and social relationships from seemingly innocuous interactions, implicit privacy is emerging as a challenge beyond data disclosure and access control. Such risks arise when AI systems derive sensitive knowledge or social inferences that individuals neither intentionally reveal nor fully anticipate.
- Our research integrates privacy engineering with computational social science to study human-centered implicit privacy. We develop frameworks for implicit privacy discovery, risk awareness, and privacy-preserving interventions, while examining how AI-driven inference shapes disclosure behaviors, privacy perceptions, social interactions, and evolving privacy norms in inference-rich digital environments.

## Publications

- **Statistics:** Google scholar citations > 542, h-index = 11; published 7 papers as first or corresponding author, including 3 in CCF-A, 1 with impact factor > 10, and 1 highly cited in ESI

• **Representative publications:** [J12], [J3], and [J1] in *IEEE Transactions on Dependable and Secure Computing*; [J7] and [J19] in *IEEE Transactions on Information Forensics and Security* ; [J15] in *ACM Computing Surveys*; [J5] in *IEEE Transactions on Pattern Analysis and Machine Intelligence*; [C2–C4] in *ACM International Conference on Multimedia*

## Refereed Journal Papers

- J1. Wen, W., Wu, Z., Zhang, Y., **Wang, Tao**, Xiao, X. & Fang, Y. Medical Archive in an Image: Generating a High-Capacity Customizable Cover Image for Medical Privacy Protection. *IEEE Transactions on Dependable and Secure Computing* **23**, 4735–4747 (2026).
- J2. Wen, W., Yuan, Z., Zhang, Y., **Wang, Tao**, Xiao, X., Zhao, R. & Fang, Y. Image Privacy Protection: A Survey. *ACM Computing Surveys* **58**, 1–36 (2026).
- J3. Xu, K., Wen, W., Qi, S., **Wang, Tao**, Zhang, Y. & Fang, Y. Detecting Malicious Concepts without Image Generation in AI-Generated Content (AIGC). *IEEE Transactions on Dependable and Secure Computing* **23**, 8699–8716 (2026).
- J4. Zhang, X., Fu, Y., **Wang, Tao**, Jiang, Z. & Zhang, M. Mobile-SST-RIR: Resource-aware face image privacy processing for IoT visual devices with key-conditioned reversible recovery. *Internet of Things* **39**, 102061 (2026).
- J5. Zhang, Y., Ji, J., **Wang, Tao**, Zhao, R., Wen, W. & Xiang, Y. Make Identity Indistinguishable: Utility-Preserving Face Dataset Publication With Provable Privacy Guarantees. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **48**, 127–139 (2026).
- J6. Li, M., Zhou, W., **Wang, Tao**, Zhang, Y. & Wen, W. A Comprehensive Image Protection Framework Based on High-Capacity Adversarial Data Hiding. *IEEE Transactions on Big Data* **11**, 3314–3329 (2025).
- J7. **Wang, Tao**, Wen, W., Xiao, X., Hua, Z., Zhang, Y. & Fang, Y. Beyond Privacy: Generating Privacy-Preserving Faces Supporting Robust Image Authentication. *IEEE Transactions on Information Forensics and Security* **20**, 2564–2576 (2025).
- J8. **Wang, Tao**, Zhang, Y., Xiao, X., Xu, K., Yuan, L., Wen, W. & Fang, Y. Make Identity Unextractable yet Perceptible: Synthesis-Based Privacy Protection for Subject Faces in Photos. *arXiv preprint arXiv:2509.11249* (2025).
- J9. **Wang, Tao**, Zhang, Y., Yang, Z., Xiao, X., Zhang, H. & Hua, Z. Seeing is Not Believing: An Identity Hider for Human Vision Privacy Protection. *IEEE Transactions on Biometrics, Behavior, and Identity Science* **7**, 170–181 (2025).
- J10. Xie, Y., Zhou, Y., **Wang, Tao**, Hua, Z., Wen, W., Yi, S. & Zhang, Y. Controllable facial protection against malicious translation-based attribute editing. *Knowledge-Based Systems* **309**, 112873 (2025).
- J11. Zhang, X., **Wang, Tao**, Ji, J., Zhang, Y. & Lan, R. Privacy-preserving face attribute classification via differential privacy. *Neurocomputing* **626**, 129556 (2025).
- J12. Zhang, Y., Yang, Z., **Wang, Tao**<sup>†</sup>, Hua, Z. & Weng, J. Tailor-Made Face Privacy Protection via Class-Wise Targeted Universal Adversarial Perturbations. *IEEE Transactions on Dependable and Secure Computing* **22**, 5108–5120 (2025).
- J13. Zhao, R., Zhang, Y., **Wang, Tao**, Wen, W., Xiang, Y. & Cao, X. Visual content privacy protection: A survey. *ACM Computing Surveys* **57**, 1–36 (2025).

- J14. Ji, J., Zhao, R., **Wang, Tao** & Zhu, Y. Utility-enhanced image obfuscation with block differential privacy. *IEEE Signal Processing Letters* **31**, 2135–2139 (2024).
- J15. **Wang, Tao**, Zhang, Y., Qi, S., Zhao, R., Xia, Z. & Weng, J. Security and Privacy on Generative Data in AIGC: A Survey. *ACM Computing Surveys* **57**, 1–34 (2024).
- J16. **Wang, Tao**, Zhang, Y., Zhao, R., Wen, W. & Zhu, Y. Protecting Face Privacy via Beautification. *Journal of Frontiers of Computer Science & Technology* **18**, 244 (2024).
- J17. Xie, Y., Zhou, Y., **Wang, Tao**†, Wen, W., Yi, S. & Zhang, Y. Reversible Gender Privacy Enhancement via Adversarial Perturbations. *Neural Networks* **172**, 106130 (2024).
- J18. **Wang, Tao**, Zhang, Y., Zhao, R., Wen, W. & Lan, R. Identifiable face privacy protection via virtual identity transformation. *IEEE Signal Processing Letters* **30**, 773–777 (2023).
- J19. Zhang, Y., **Wang, Tao**, Zhao, R., Wen, W. & Zhu, Y. RAPP: Reversible Privacy Preservation for Various Face Attributes. *IEEE Transactions on Information Forensics and Security* **18**, 3074–3087 (2023).
- J20. Sun, L., Liu, X., Liu, Y., **Wang, Tao**, Guo, L., Zheng, X. & Luo, Y. A novel deep recommend model based on rating matrix and item attributes. *Journal of Intelligent Information Systems* **57**, 295–319 (2021).

## Refereed Conference Papers

- C1. **Wang, Tao**. *Information-Theoretic Face Anonymization in Under Review* (2026).
- C2. Xu, K., Zhang, Y., **Wang, Tao**, Qi, S., Carminati, B., Ferrari, E. & Fang, Y. *Concept-Level Risk and Calibration for Governance in Diffusion Foundation Models in Proceedings of the ACM International Conference on Multimedia* (2026), Just Accepted.
- C3. **Wang, Tao**, Zhang, Y., Xiao, X., Yuan, L., Xia, Z. & Weng, J. *Make Privacy Renewable! Generating Privacy-Preserving Faces Supporting Cancelable Biometric Recognition in Proceedings of the ACM International Conference on Multimedia* (2024), 10268–10276.
- C4. Yang, Z., Zhang, Y., **Wang, Tao**, Hua, Z., Xia, Z. & Weng, J. *Once-for-all: Efficient Visual Face Privacy Protection via Person-Specific Veils in Proceedings of the ACM International Conference on Multimedia* (2024), 7705–7713.

## Awards & Honors

- Doctoral Student Program of the Young Sci-Tech Talents Cultivation Project, CAST, 2025
- Puxin Elite Scholarship, 2025
- CSC Scholarship of High-Level Postgraduate Program, 2025
- ACM MM Honorable Mention Award, 2024
- National Scholarships for Postgraduate Students, 2023

## Research Grants

- **PI**, Postgraduate Research Innovation Program, Jiangsu Provincial Department of Education; Friendly Face Privacy Protection      CN¥ 15,000      2025–2027

## Presentations & Talks

2026 **MILAN0DAY**, University of Milan. Invited Talk: “AI and the Death of Privacy? Navigating New Threats and Modern Defenses.”

## Professional Services

**Journal Reviewers:** TKDE, TDSC, TCSVT, TNSE, TBIOM, TOMM, EAAI, ESWA, KBS, IOTJ.

**Conference Reviewer:** ICLR, SIGKDD, AAAI, ACM MM

**Membership:** ACM EIGTRUST Membership; CSIG Membership

## References

- **Pierangela Samarati**, General supervisor, pierangela.samarati@unimi.it , University of Milan
- **Yushu Zhang**, Ph.D. supervisor, zhangyushu@jxufe.edu.cn, Jiangxi University of Finance and Economics
- **Yuming Fang**, General supervisor, leo.fangyuming@foxmail.com, , Jiangxi University of Finance and Economics
- **Shuren Qi**, General supervisor, shurenqi@nuaa.edu.cn, City University of Hong Kong

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